AI ENTERTAINMENT SYSTEM

A Project-II Report

Submitted in partial fulfillment of requirement of the

Degree of

BACHELOR OF TECHNOLOGY in COMPUTER SCIENCE & ENGINEERING

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JAN-JUNE 2026

Report Approval

The project work **"AI Entertainment System"** is hereby approved as a creditable study of an engineering/computer application subject carried out and presented in a manner satisfactory to warrant its acceptance as prerequisite for the Degree for which it has been submitted.

It is to be understood that by this approval the undersigned do not endorse or approve any statement made, opinion expressed, or conclusion drawn therein; but approve the "Project Report" only for the purpose for which it has been submitted.

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Declaration

We hereby declare that the project entitled **"AI Entertainment System"** submitted in partial fulfillment for the award of the degree of Bachelor of Technology/Master of Computer Applications in 'Engineering' completed under the supervision of **Prof. Ajeet Singh Rajput, Prof. Shyam Patel and Prof. (Dr.) Hemlata Patel,** Department of Computer Science & Engineering, Faculty of Engineering, Medicaps University Indore is an authentic work.

Further, we declare that the content of this Project work, in full or in parts, has neither been taken from any other source nor submitted to any other Institute or University for the award of any degree or diploma.

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Certificate

We, **Prof. Ajeet Singh Rajput , Prof. Shyam Patel and Prof. (Dr.) Hemlata Patel** certify that the project entitled "**AI Entertainment System**" submitted in partial fulfillment for the award of the degree of Bachelor of Technology by **Prabal Shukla, Pratik Koushal, Priyansh Chouhan, Pushpraj Singh Chouhan & Rajat Malviya** is the work carried out by them under our guidance and that the work has not formed the basis of award of any other degree elsewhere.

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Abstract

The **AI Entertainment System** is a full-stack intelligent web application designed to deliver interactive and personalized entertainment solutions by integrating **Agentic AI, Generative AI, and DevOps practices**. The system combines a modern frontend developed using React and Tailwind CSS with a high-performance backend built using FastAPI.

The system leverages Generative AI (GenAI) through the Groq API and Large Language Models (LLMs) to generate human-like responses for tasks such as movie recommendations, entertainment suggestions, and script writing. In addition, an Agentic AI layer is incorporated to manage task flow by understanding user intent, planning actions, invoking AI services, and refining responses, thereby enabling semi-autonomous decision-making within the system.

To enhance contextual understanding, the project explores Retrieval-Augmented Generation (RAG) using vector databases such as FAISS, allowing the system to retrieve relevant information and improve response accuracy.

Furthermore, DevOps practices are applied using containerization and deployment strategies to ensure scalability, reliability, and efficient system management. The integration of these technologies results in a modular, scalable, and intelligent system capable of delivering fast, relevant, and structured responses.

The results demonstrate that the system effectively combines Agentic AI for orchestration, Generative AI for content creation, and DevOps for deployment, making it suitable for real-world AI-driven entertainment applications.

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Abbreviations

Abbreviation	Full Form
AI	Artificial Intelligence
NLP	Natural Language Processing
LLM	Large Language Model
RAG	Retrieval-Augmented Generation
API	Application Programming Interface
UI	User Interface
UX	User Experience
DB	Database
JSON	JavaScript Object Notation
HTTP	HyperText Transfer Protocol
CI/CD	Continuous Integration / Continuous Deployment
DevOps	Development and Operations
Agentic AI	AI systems capable of decision-making and task planning
Gen AI	AI systems that create new content (text, images, audio, or code) by learning patterns from data.

Notations & Symbols

Symbol	Meaning
→	Data Flow
↔	Bidirectional Communication
API	Communication Interface
LLM	AI Processing Unit
DB	Data Storage
[]	Array / List
{}	JSON Object
()	Function Call
AI	Artificial Intelligence Component
Agent	Decision-making unit (Agentic AI)
Cloud	Deployment environment (DevOps)

Chapter 1

Introduction

1.1 Introduction

With the rapid advancement of Artificial Intelligence (AI) and Natural Language Processing (NLP), modern applications are increasingly integrating intelligent systems to enhance user interaction and experience. AI-driven systems are now capable of understanding human language, generating meaningful responses, and providing personalized recommendations across various domains.

The AI Entertainment System is designed as a full-stack web application that leverages AI to provide intelligent entertainment solutions. The system enables users to interact with an AI-powered chatbot that can generate movie recommendations, entertainment suggestions, and even creative content such as scripts.

The application integrates a React-based frontend with a FastAPI backend, and utilizes a Large Language Model (LLM) via the Groq API for generating intelligent responses (**Generative AI**). In addition, the system incorporates an **Agentic AI approach**, where the system processes user intent, plans actions, and refines responses dynamically to improve output quality. The system is designed to be scalable, modular, and user-friendly, ensuring a seamless interaction experience.

Additionally, the system explores the concept of Retrieval-Augmented Generation (RAG), where external knowledge can be integrated to enhance the accuracy and contextual relevance of AI-generated responses. Furthermore, basic **DevOps practices such as containerization and deployment planning are considered to ensure system scalability and maintainability.**

1.2 Literature Review

Recent developments in AI and NLP have significantly improved the capability of machines to understand and generate human-like text.

• The Transformer architecture, introduced by Ashish Vaswani et al., revolutionized NLP by enabling parallel processing and attention mechanisms.
• GPT-based models have demonstrated strong capabilities in conversational AI, content generation, and reasoning tasks (**Generative AI systems**).
• Research on Retrieval-Augmented Generation (RAG) combines retrieval systems with generative models to improve factual accuracy and reduce hallucinations.
• Modern APIs such as Groq LLM services provide high-speed inference, enabling real-time AI applications.
• Recent studies on **Agentic AI systems** highlight the importance of autonomous decision-making, planning, and task orchestration in intelligent applications.

These advancements have made it possible to build intelligent applications like the AI Entertainment System.

1.3 Objectives

The primary objectives of this project are:

• To develop a full-stack AI-powered web application
• To implement a real-time chat system using AI
• To provide personalized entertainment recommendations
• To integrate Large Language Models (LLMs) into web applications (**GenAI**)
• To incorporate **Agentic AI for intelligent task handling and response refinement**
• To design a scalable and modular architecture
• To explore RAG-based enhancements for improved response accuracy
• To apply basic **DevOps practices for deployment and scalability**

1.4 Significance

The AI Entertainment System demonstrates the practical application of AI in enhancing user engagement and personalization.

Key significance includes:

• Provides an interactive and intelligent user experience
• Demonstrates real-world integration of LLMs (**Generative AI**)
• Introduces **Agentic AI concepts for intelligent workflow handling**
• Serves as a foundation for AI-based recommendation systems
• Enhances understanding of full-stack AI system design
• Bridges the gap between theoretical AI concepts and practical implementation
• Highlights the role of **DevOps in deploying scalable AI systems**

1.5 Research Design

The system follows a modular and layered architecture, ensuring separation of concerns and scalability.

Design Components:

• Frontend Layer
o Developed using React and Tailwind CSS
o Handles user interaction and UI rendering

• Backend Layer
o Developed using FastAPI
o Handles API requests, authentication, and processing

• AI Layer (**Generative AI**)
o Uses Groq API for LLM-based response generation

• Agentic Layer
o Handles user intent understanding, planning, and response refinement

• Optional RAG Layer
o Uses FAISS and embeddings for contextual retrieval

• DevOps Layer
o Supports deployment, containerization, and scalability

1.6 Source of Data

The system utilizes multiple data sources:

- **User Input Data**
  - Queries entered by users through the frontend
- **AI Generated Data**
  - Responses generated by the LLM
- **API Data**
  - Communication between frontend and backend
- **Optional External Data (RAG)**
  - Retrieved documents from vector database

1.7 Chapter Scheme

- Chapter 1: Introduction
- Chapter 2: System Design & Implementation
- Chapter 3: Working of System
- Chapter 4: Implementation Details
- Chapter 5: Results and Discussion
- Chapter 6: Conclusion
- Chapter 7: Future Scope

Chapter 2

System Design & Implementation

2.1 Experimental Setup

The AI Entertainment System is developed using a full-stack architecture integrating frontend, backend, and AI services.

Hardware Requirements

• Processor: Intel i3 / Ryzen 3 or above
• RAM: Minimum 4 GB (8 GB recommended)
• Storage: 5 GB free space
• Internet: Required for API calls

Software Requirements

• Operating System: Windows / Linux / macOS
• IDE: Visual Studio Code
• Runtime: Node.js (Frontend), Python 3.10+ (Backend)
• Browser: Google Chrome / Edge

Development Environment

• Frontend: React (Vite) + Tailwind CSS
• Backend: FastAPI (Python)
• AI Integration: Groq LLM API (**Generative AI**)
• Optional GenAI Enhancement: RAG using FAISS + embeddings
• Agentic AI Layer: Handles user intent processing, response planning, and dynamic refinement
• DevOps: Docker (for containerization), GitHub (version control), optional cloud deployment (AWS)

2.2 Procedures Adopted

The development of the system followed a structured approach:

1. Requirement Analysis

• Identified need for AI-based entertainment system
• Defined features: login, chat, dashboard
• Identified need for intelligent AI interaction (**Agentic AI concept**)

2. System Design

• Designed layered architecture
• Planned API structure
• Defined AI workflow and response pipeline

3. Frontend Development

• Built UI using React and Tailwind
• Implemented routing and state management
• Designed user-friendly dashboard and chat interface

4. Backend Development

• Created REST APIs using FastAPI
• Integrated authentication and chat endpoints
• Handled request processing and API communication

5. AI Integration (GenAI + Agentic AI)

• Connected Groq LLM for response generation (**Generative AI**)
• Implemented prompt-based interaction
• Applied **Agentic AI logic for understanding user intent and refining outputs**

6. Testing

• API testing using browser and Swagger
• UI testing for responsiveness
• Verified chat functionality and AI responses

7. DevOps Implementation

• Containerization using Docker
• Version control using Git and GitHub
• Preparation for deployment on cloud platforms (AWS)

2.3 System Architecture

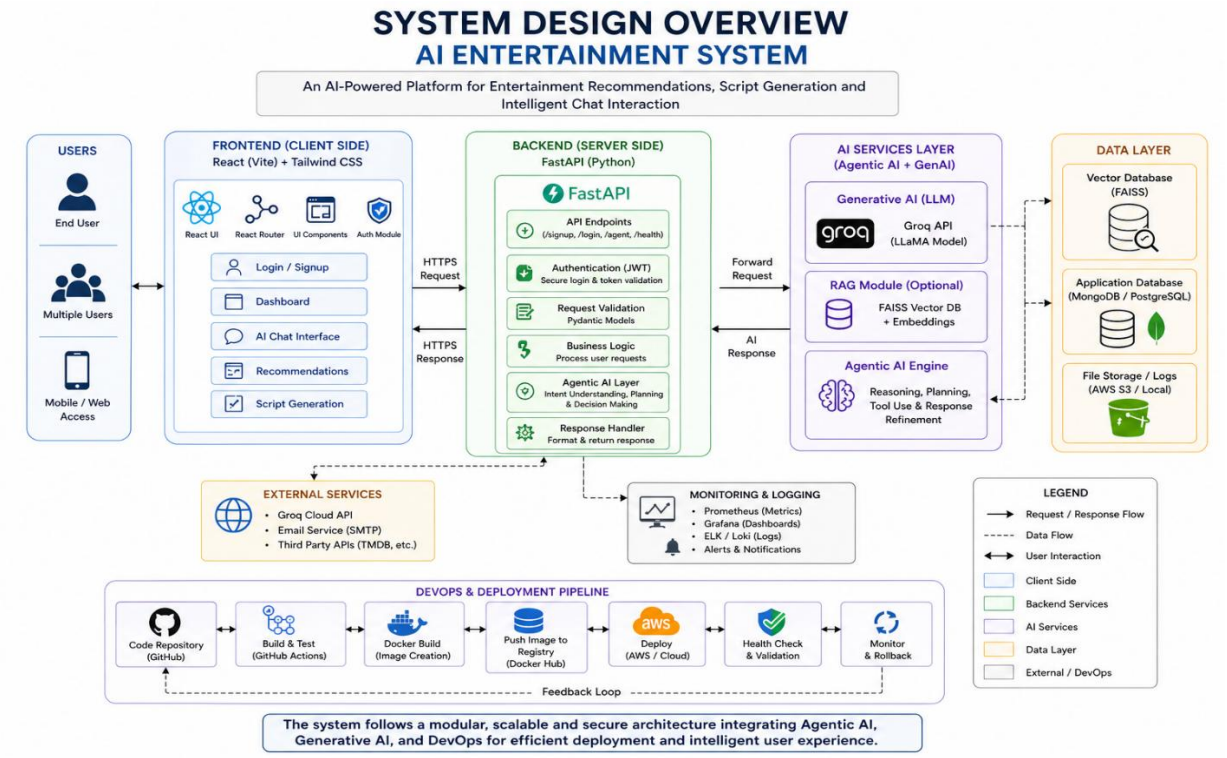


Fig 1: System Design Overview of AI Entertainment System

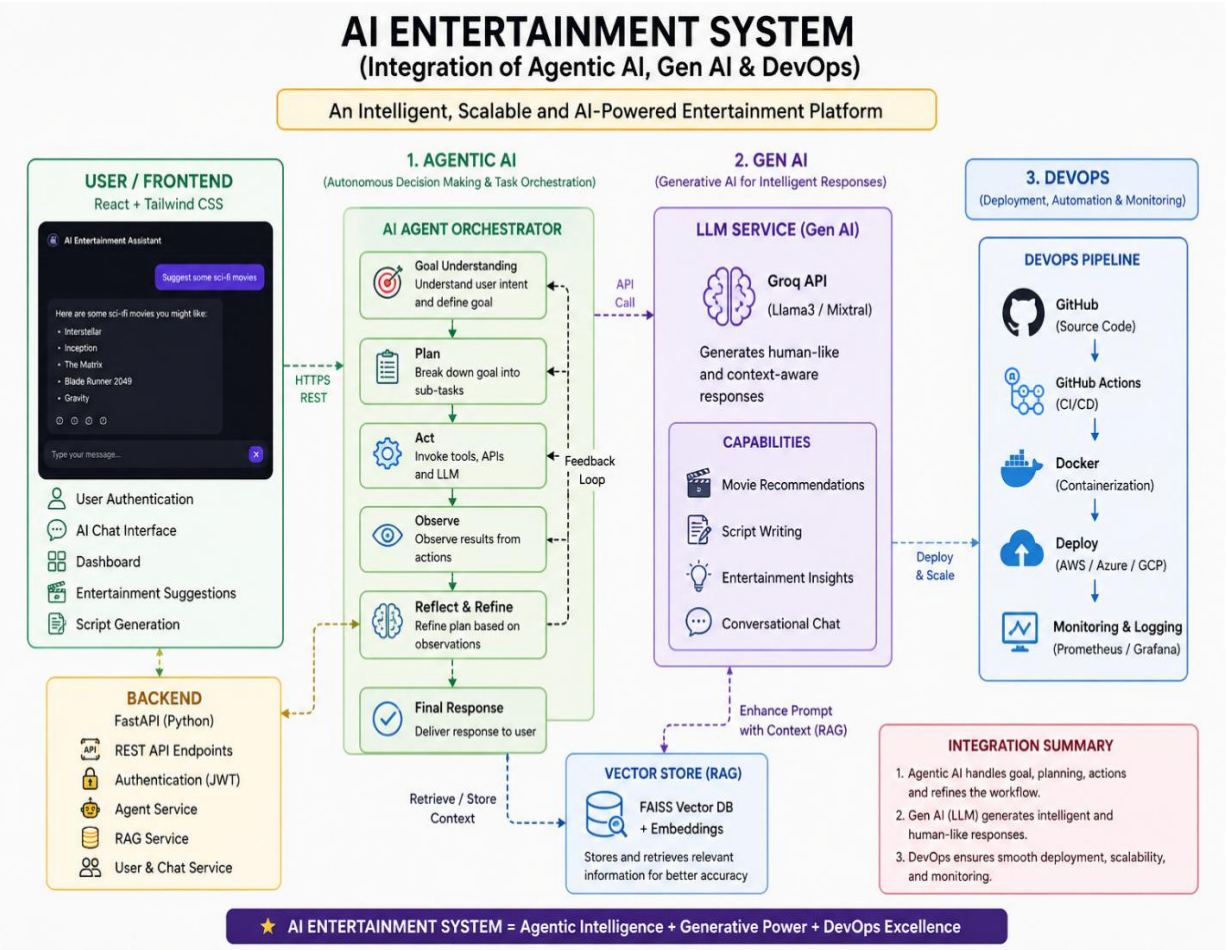


Fig 2.1: Integrated Architecture of AI Entertainment System

UML DIAGRAMS

2.4 Use Case Diagram

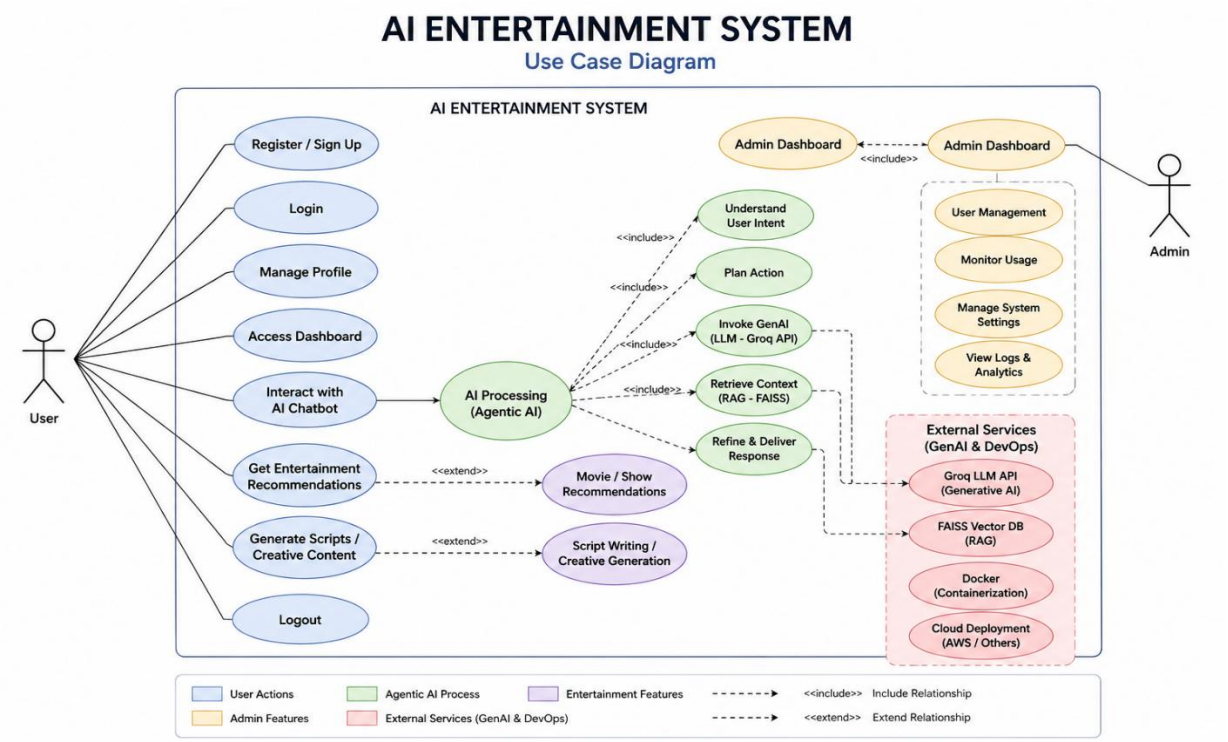


Fig 2.2: Use Case Diagram of AI Entertainment System (Integrated with Agentic AI, GenAI & DevOps)

2.5 Sequence Diagram

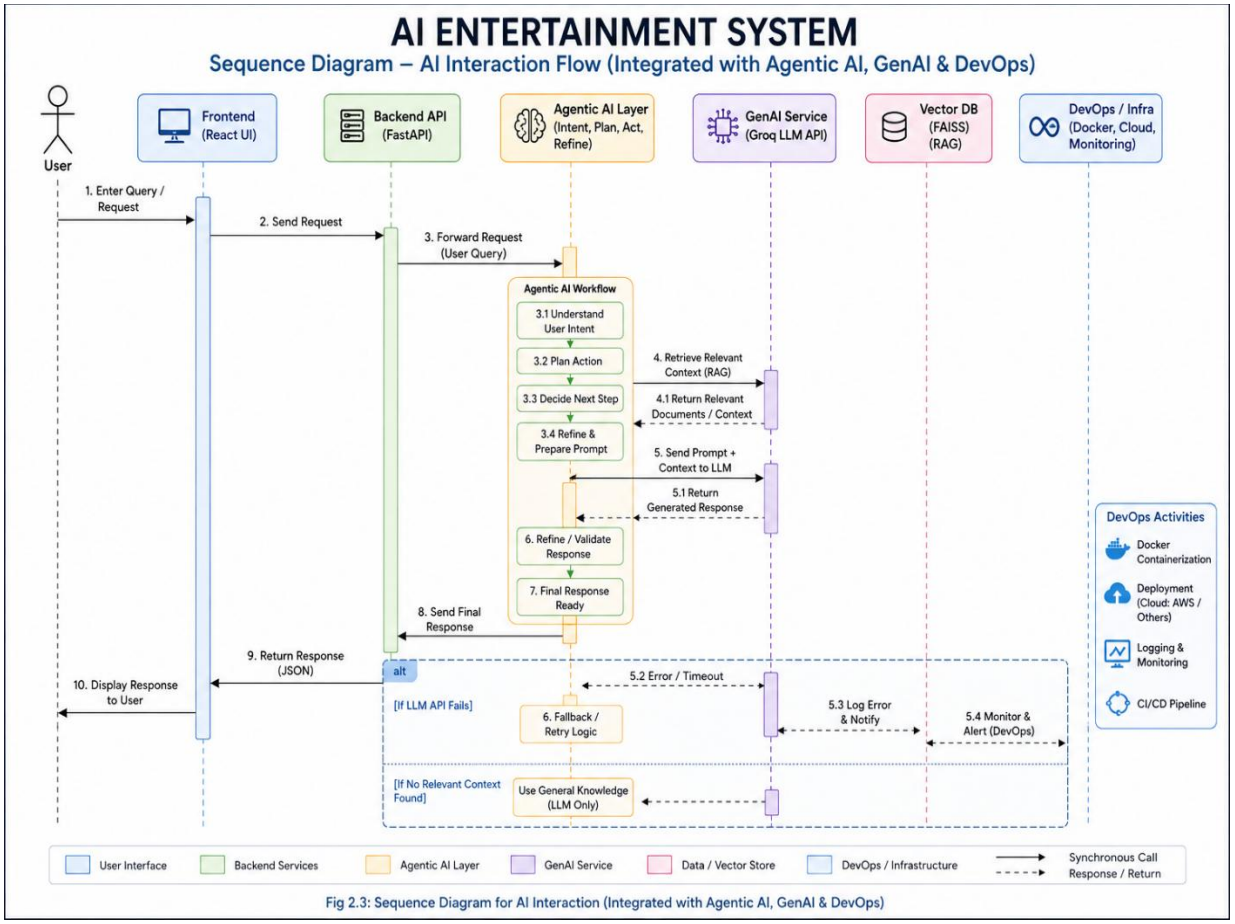


Fig 2.3: Sequence Diagram for AI Interaction (Integrated with Agentic AI, GenAI & DevOps)

2.6 Process Flow Diagram

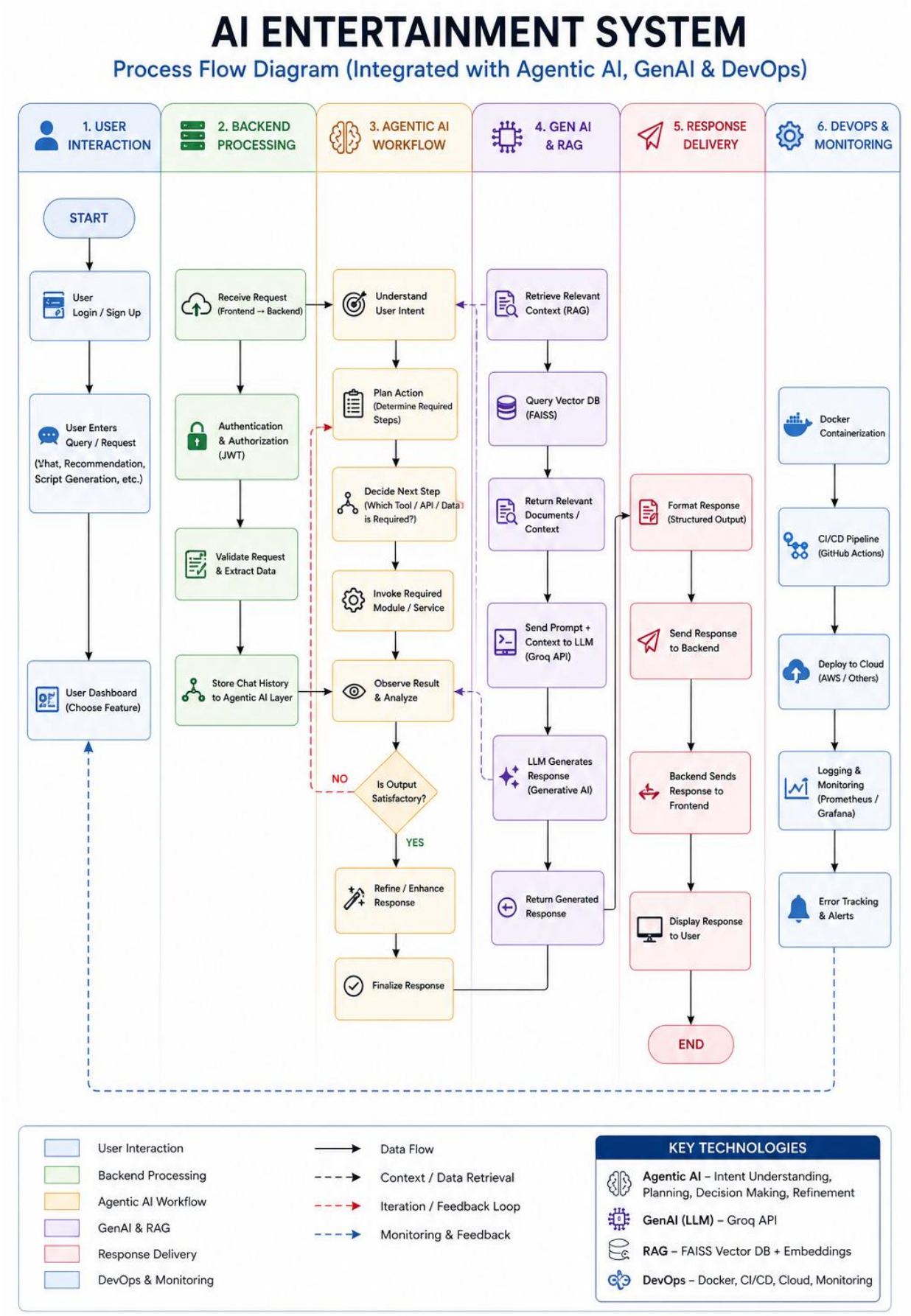


Fig 2.4: Process Flow Diagram of AI Entertainment System (Integrated with Agentic AI, GenAI & DevOps)

2.7 Class Diagram

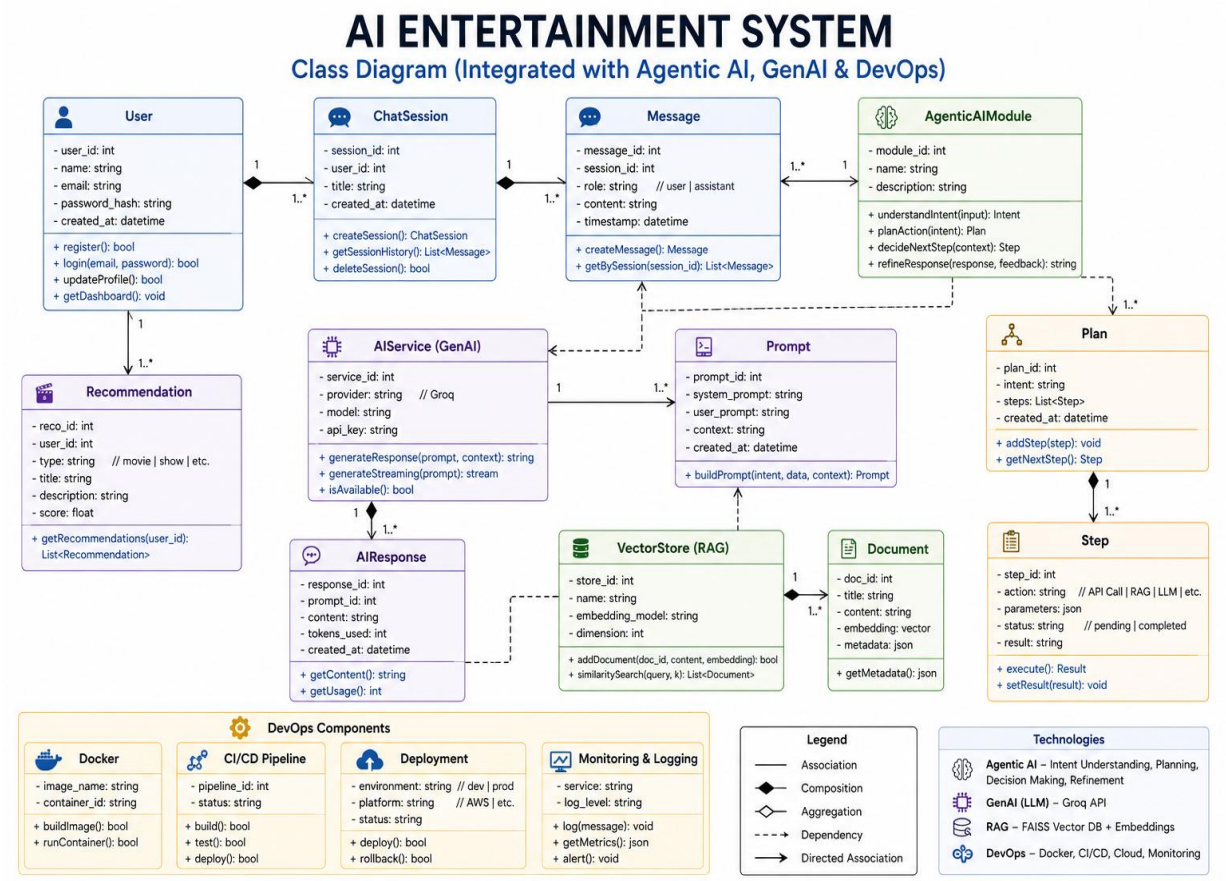


Fig 2.5: Class Diagram of AI Entertainment System (Integrated with Agentic AI, GenAI & DevOps)

2.8 Deployment Diagram

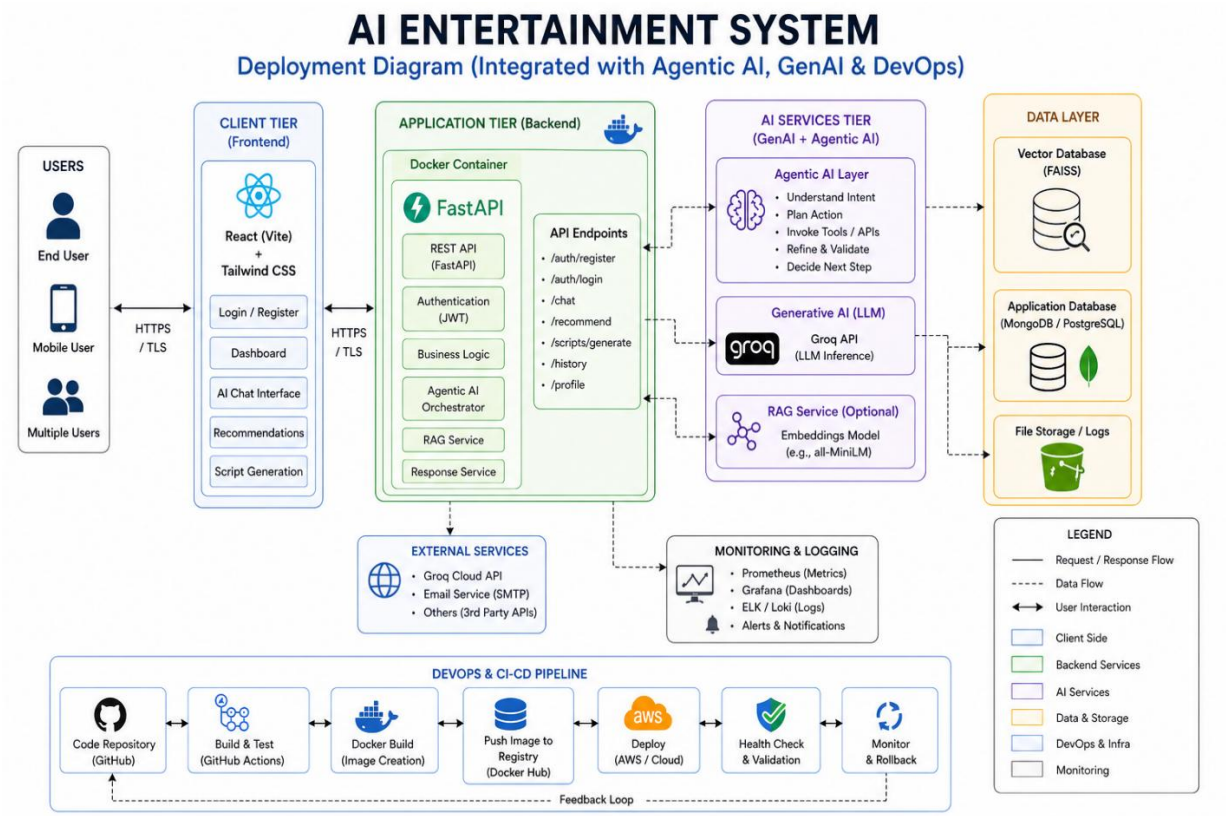


Fig 2.6: Deployment Diagram of AI Entertainment System (Integrated with Agentic AI, GenAI & DevOps)

Chapter 3

Working of System

3.1 Overview of System Workflow

The AI Entertainment System operates as a full-stack intelligent application where user interactions are processed through multiple layers, including frontend, backend, and AI services.

The workflow is designed to ensure:

• Real-time interaction
• AI-based response generation (**Generative AI**)
• Scalable architecture
• Modular processing
• Intelligent decision-making using **Agentic AI**

Overall Flow

- User interacts with frontend (React UI)
- Request is sent to backend (FastAPI)
- Backend processes and forwards request to **Agentic AI layer**
- Agentic AI refines request and interacts with AI model
- AI generates response (via Groq LLM – GenAI)
- Response is returned and displayed to user

3.2 Frontend Workflow (React Application)

The frontend is built using **React (Vite + Tailwind CSS)** and is responsible for user interaction.

Working Steps:

- User logs in or signs up
- Dashboard is loaded after authentication
- User navigates to chat interface
- User enters a query (e.g., “movie recommendations”)
- API request is sent to backend using fetch/axios

Key Functionalities:

- Authentication UI (Login/Signup)
- Dashboard with stats and recent chats
- Chat interface for AI interaction

- Navigation using React Router

Frontend Flow:

User → Login → Dashboard → Chat → Send Message → API Call

3.3 Backend API Workflow (FastAPI Server)

The backend is developed using FastAPI and acts as the core processing unit.

Working Steps:

- API receives request (/agent)
- Request is validated using Pydantic models
- Authentication token is verified
- Backend forwards request to **Agentic AI processing layer**
- Agentic AI prepares structured input
- Request forwarded to AI service (Groq API)
- Response processed and returned to frontend

Key Endpoints:

• /signup → Register user
• /login → Authenticate user
• /agent → AI chat processing
• /health → Server health check

Backend Flow:

Frontend Request → API Route → Validation → Processing → AI

3.4 AI Processing Flow (Agentic AI + GenAI Integration)

The AI layer is responsible for generating intelligent responses using Groq LLM (LLaMA model).

Working Steps:

- Backend sends user query to Agentic AI module
- Agentic AI understands user intent
- Agentic AI plans response generation
- Prompt is constructed (Prompt Engineering)

- AI processes the input (Generative AI)
- Response is generated
- Agentic AI refines response (if needed)
- Response returned to backend

Example:

User Input:
"Suggest movies"

AI Output:
Here are some movie recommendations:

1. Inception
2. Interstellar
3. The Dark Knight
...

3.5 RAG (Retrieval-Augmented Generation) – Partial Implementation

The system includes a **partially implemented RAG module** using:

- FAISS (Vector Database)
- Sentence Transformers (Embeddings)

Working Steps:

- User query is converted into vector
- Similar documents retrieved from FAISS
- Context is added to prompt
- AI generates response using context

RAG Flow:

User Query → Embedding → Vector Search (FAISS) → Context → LLM → Response

3.6 Authentication Flow

The system uses **token-based authentication**.

Steps:

- User signs up or logs in

- Backend generates token
- Token stored in localStorage
- Token sent with every request
- Backend validates token

Flow:

Login → Token Generated → Stored → Used in API Calls → Verified

3.7 Data Flow

The data flow in the system is continuous and bidirectional:

• User → Frontend → Backend → Agentic AI → GenAI → Backend → Frontend → User

Each layer processes and transforms data to ensure correctness and usability.

3.8 Error Handling & System Behavior

Frontend:

• Displays loading state
• Handles API failures
• Redirects unauthorized users

Backend:

• Returns proper HTTP status codes
• Handles exceptions
• Logs errors

DevOps Layer:

• Logging and monitoring of system performance
• Error tracking and alerting
• CI/CD support for updates

3.9 Security Considerations

• JWT-based authentication
• Secure API communication (HTTPS)
• Input validation to prevent misuse
• Secure deployment using DevOps practices

Chapter 4

Implementation Details

4.1 Overview of Implementation

The AI Entertainment System is implemented using a modular approach, where each functional component is developed independently and integrated through well-defined APIs. The system follows a client-server architecture with clear separation between frontend, backend, and AI layers.

Additionally, the system integrates:

• **Generative AI (LLM)** for response generation
• **Agentic AI** for intelligent decision-making and prompt handling
• **DevOps practices** for deployment and scalability

4.2 Authentication Module

The authentication module ensures secure access to the system and manages user sessions.

Features

• User registration (signup)
• User login (authentication)
• Token-based session management

Implementation Details

• User credentials are collected via frontend forms
• Backend validates credentials using secure logic
• Upon successful login, a JWT (JSON Web Token) is generated
• Token is stored on client-side and sent with each request

Security Measures

• Passwords are stored in encrypted/hashed format
• Unauthorized access is restricted
• Token validation is performed on each API call

4.3 Chat / Agent Module

This module is the core component responsible for user interaction with the AI system.

Functionality

• Accepts user queries
• Sends queries to backend
• Uses **Agentic AI logic** to process user intent
• Displays AI-generated responses

Working

1. User enters input in chat UI
2. Request sent to backend /agent endpoint
3. Backend passes query to **Agentic AI layer**
4. Agent decides how to process the query
5. Query forwarded to **GenAI (LLM)**
6. Response returned and displayed

Features

• Real-time conversational interface
• Context-aware responses
• Multi-purpose usage (recommendations, scripts, etc.)
• Intelligent handling via **Agentic AI**

4.4 Dashboard Module

The dashboard provides an overview of user activity and system interactions.

Features

• Displays recent chat history
• Shows basic statistics
• Provides quick access to chat system

Implementation

• Data fetched from backend APIs
• Rendered dynamically using React components

4.5 GenAI Module (LLM Integration)

This module integrates the system with the Groq API for AI processing.

Key Responsibilities

• Constructing prompts
• Sending requests to LLM
• Processing AI responses

Implementation Steps

1. User input received
2. Input passed to **Agentic AI layer**
3. Agent refines and structures prompt
4. Request sent to Groq API (LLM)
5. Response received and formatted

4.6 DevOps Module (Docker Setup)

The system uses DevOps practices for deployment and scalability.

Technologies Used

• Docker (Containerization)
• Optional CI/CD pipelines (GitHub Actions)
• Local development environment

Benefits

• Platform independence
• Easy deployment
• Scalable architecture
• Faster development and testing

Implementation

• Dockerfile created for backend
• Application runs inside container
• Dependencies managed efficiently
• Environment variables handled securely

4.7 Database Handling

• Stores user data and chat history
• Ensures data persistence
• Supports future scalability

RAG Integration

• Vector database (FAISS) used
• Stores embeddings of data
• Retrieves context for better AI responses

Chapter 5

Results and Discussion

5.1 Overview

This chapter presents the results obtained from the implementation of the AI Entertainment System. It evaluates system performance, output quality, and user interaction.

The results highlight the integration of:

• **Generative AI (LLM)** for intelligent response generation
• **Agentic AI** for decision-making and prompt handling
• **DevOps practices** for stable execution and deployment

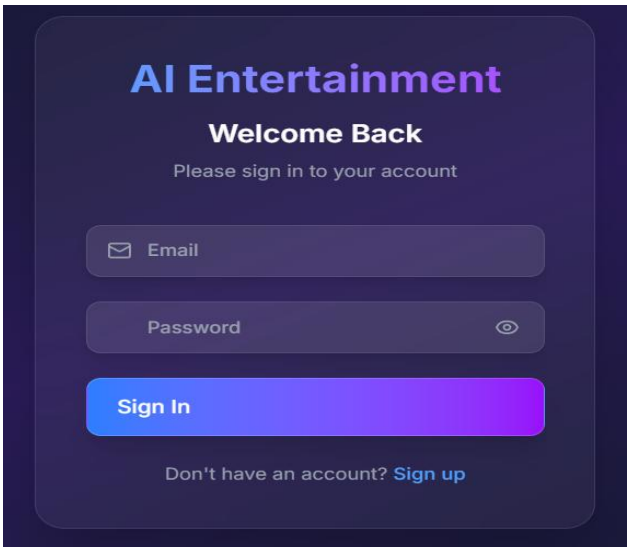
5.2 Functional Results

The system successfully performs the following functions:

• User authentication (Login/Signup)
• AI-based chat interaction
• Entertainment recommendation generation
• Script writing capability
• Dashboard display with user information

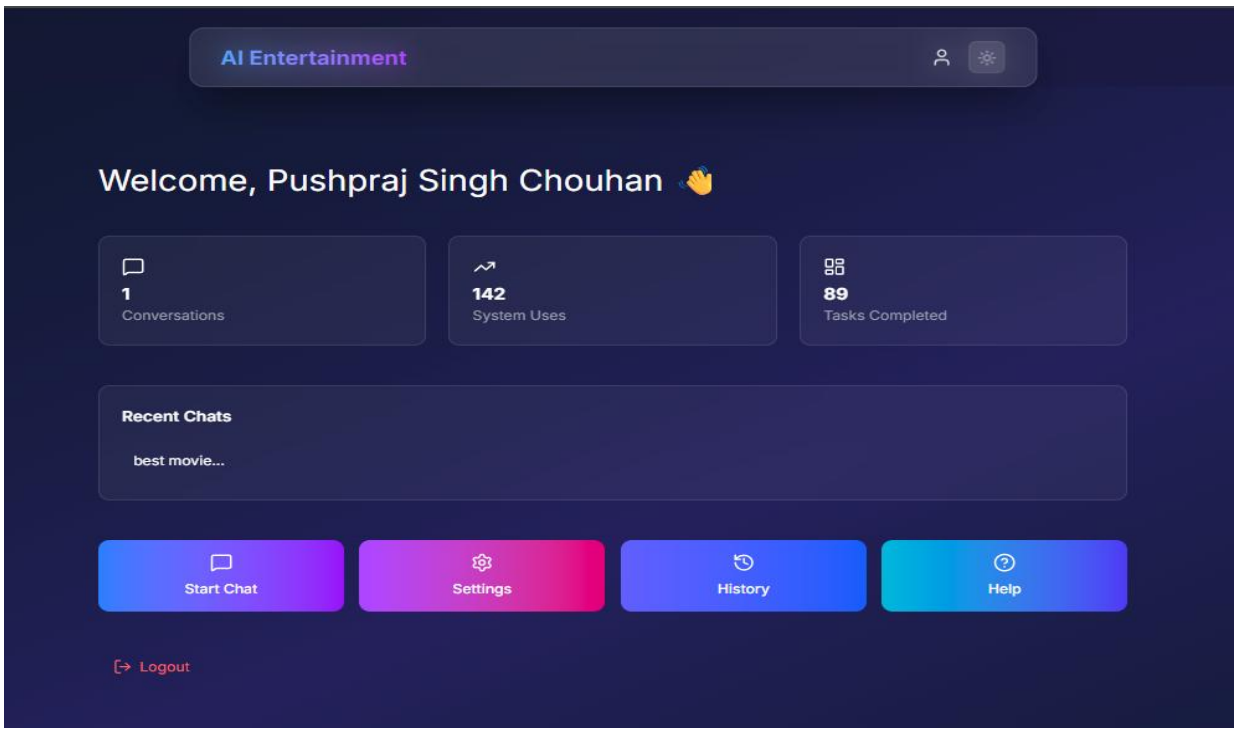
5.3 Output Screenshots

Figure 5.1: Login Page Interface



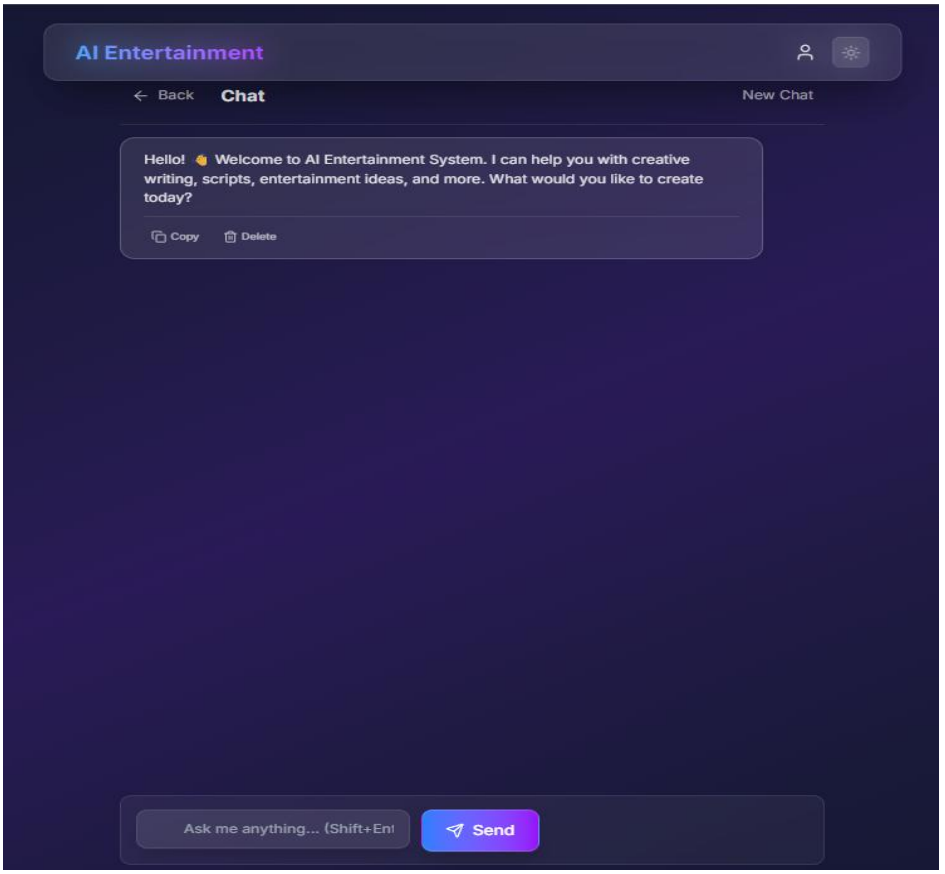
Description:
This screen allows users to enter their credentials and access the system securely.

Figure 5.2: Dashboard Interface



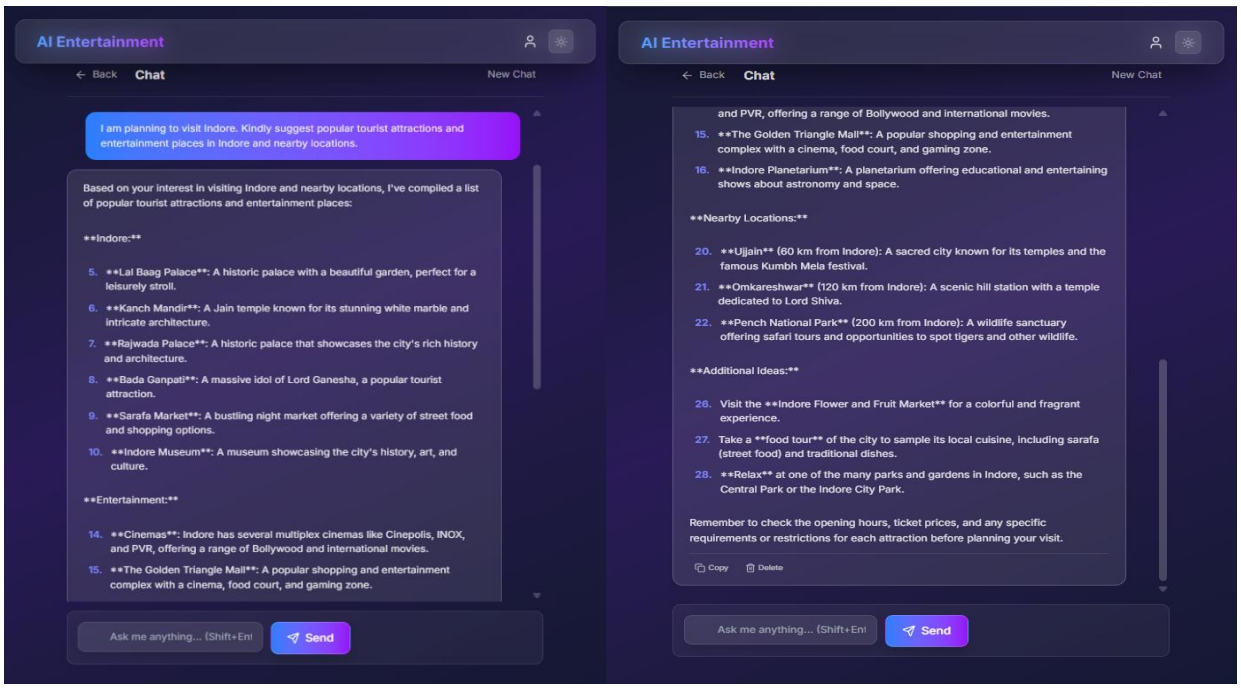
Description:
Displays user details, statistics, and navigation options such as chat , settings , help and history.

Figure 5.3: Chat Interface



Description:
Provides an interactive interface for users to communicate with the AI system.

Figure 5.4: AI Generated Output



Description:
Shows AI-generated responses such as entertainment recommendations and script generation.

5.4 System Performance

The system performs efficiently in terms of:

• Fast response generation using Groq API
• Low latency in frontend-backend communication
• Smooth UI interaction
• Stable API performance

5.5 Discussion

The system demonstrates effective integration of AI into a web application. The chatbot provides meaningful and structured responses. However, the system can be improved further using:

Key Observations:

• **Generative AI** produces relevant and human-like responses
• **Agentic AI** improves response structure and intent handling
• System architecture supports scalability
• Frontend and backend integration works efficiently

5.6 Limitations: • Dependent on internet connectivity
• AI response time may vary
• Partial implementation of RAG
• Limited long-term memory (no full vector DB yet)

Chapter 6

Conclusion

6.1 Conclusion

The AI Entertainment System successfully integrates modern web technologies with Artificial Intelligence to provide an intelligent and interactive platform for users.

The system demonstrates:

• Effective full-stack development
• Real-time AI interaction using Generative AI (LLM)
• Intelligent decision-making using Agentic AI
• Scalable and modular architecture
• Deployment readiness using DevOps practices

The integration of Groq LLM enables the system to generate dynamic and meaningful responses, while the Agentic AI layer enhances how user queries are interpreted and processed. This makes the system suitable for real-world entertainment-based applications.

6.2 Key Achievements

• Developed a complete full-stack AI application
• Integrated **Generative AI (LLM)** for real-time response generation
• Implemented **Agentic AI-based chat processing**
• Implemented authentication and chat system
• Designed responsive UI
• Explored RAG-based enhancements
• Applied basic **DevOps concepts for deployment readiness**

6.3 Learning Outcomes

Through this project, the following skills were developed:

• Full-stack web development
• API design and integration
• Generative AI (LLM integration)
• Understanding of **Agentic AI concepts**
• System design and architecture
• Debugging and deployment
• Basic **DevOps and containerization concepts**

6.4 Final Statement

The project successfully achieves its objective of creating an AI-powered entertainment platform by integrating **Agentic AI, Generative AI, and DevOps practices**. It provides a strong foundation for future enhancements such as full RAG implementation, advanced personalization, and scalable cloud deployment.

Chapter 7

Future Scope

The current system provides a strong base, but several enhancements can further improve its functionality and performance.

7.1 Full RAG Implementation

• Integrate complete vector database
• Improve contextual understanding
• Enhance response accuracy

7.2 Personalization System

• Track user preferences
• Provide customized recommendations
• Use user history for better AI output

7.3 Cloud Deployment

• Deploy system on AWS or Azure
• Improve scalability and availability
• Enable global access
• Implement CI/CD pipelines using **DevOps practices**

7.4 Voice-Based Interaction

• Integrate speech-to-text
• Enable voice commands
• Improve accessibility

7.5 Mobile Application

• Develop Android/iOS app
• Improve usability and reach

7.6 Advanced AI Features

• Multi-agent systems (**Agentic AI extension**)
• Emotion-aware responses
• Real-time learning capabilities
• Integration of advanced **Generative AI models**

Appendix

Appendix A: Sample API Requests and Responses

A.1 Signup Request

```
POST /signup
{ "name": "Test User",
  "email": "test@example.com",
  "password": "123456" }
```

Response

```
{  "success": true,
  "user_id": 1,
  "token": "dummy_token_123" }
```

A.2 Login Request

```
POST /login
{  "email": "test@example.com",
  "password": "123456" }
```

Response

```
{  "success": true,
  "user_id": 1,
  "token": "dummy_token_123" }
```

A.3 Chat Request (Agentic AI + GenAI)

```
POST /agent
{  "message": "Suggest movies" }
```

Response

```
{  "response": "Here are some movie recommendations..." }
```

Appendix B: Project Folder Structure

```
AI_Entertainment_System/
│
├── frontend/
│   ├── src/
│   ├── components/
│   ├── pages/
│   └── services/
│
├── backend/
│   ├── main.py
```

```
│   ├── routes.py
│   ├── services.py
│   ├── config.py
│   └── genai/
│       └── rag.py
│
├── docker-compose.yml   (DevOps setup)
├── Dockerfile           (Containerization)
└── README.md
```

Appendix C: Key Technologies Used

• React (Frontend UI)
• FastAPI (Backend Framework)
• Groq API (LLM Integration – **Generative AI**)
• Agentic AI Logic (for decision-making and response control)
• FAISS (Vector Database - Optional RAG)
• Docker (DevOps - Containerization)

Appendix D: Environment Configuration

Example .env file:

```
GROQ_API_KEY=your_api_key_here
GROQ_MODEL=llama-3.1-8b-instant

# Backend Config
PORT=8001
DEBUG=True

# CORS
ALLOWED_ORIGINS=*
```

Appendix E: Important Code Snippets

AI Response Function (GenAI + Agentic AI)

```
async def get_ai_response(user_input):
    # Agentic AI step: process intent
    processed_input = f"User Query: {user_input}"

    # GenAI step: generate response
    return f"AI response for: {processed_input}"
```

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